

ally consists of a gas diffusion layer that absorbs oxygen from the air, a catalyst layer that enhances the ORR (oxygen reduction reaction) and OER (oxygen evolution reaction), and a current collector that connects the battery to external electrical circuits to create a closed circuit. The schematic arrangement of the Fe-air battery and its electrochemical reaction is shown in Fig. 4.

Attractive features

Metallic iron can be used directly as an anode in aqueous electrolytes due to its appropriate redox potential of -0.44V vs SHE (standard hydrogen electrode) in an acidic solution and -0.88V vs SHE in alkaline electrolyte. Redox potential is a measure of how easily a substance loses or gains electrons in a reduc-

tion-oxidation reaction. Aqueous Zn-ion batteries are one of the most attractive battery systems. However, as the redox potential of Fe is higher than that of Zn, it exhibits better stability in aqueous electrolyte than Zn. The theoretical specific capacity and volumetric capacity of Fe are about 960mAh g⁻¹ (820mAh g⁻¹ for Zn) and 7557mAh cm⁻³ (5851mAh cm⁻³ for Zn), respectively.

These properties enable Fe-air batteries to have a higher specific capacity than Zn-ion batteries. The price of iron is 1/43 that of zinc. Unlike other batteries, iron is less prone to forming dendrites with repeated charge-discharge cycles, which can severely disfigure the electrode shape.

Compared to iron-nickel batteries, the weight of this battery can be decreased by one third, as it uses a lightweight bifunctional air-breathing electrode instead of a bulky positive nickel electrode. This step increases its specific energy by up to 100%. Iron is also a very low-cost metal, although the strong alkaline electrolyte requires necessary safety measures. Another advantage is that it has a good energy density of about 600Wh kg⁻¹ with a round trip efficiency of about 90%, comparable to a Li-ion battery. Its estimated manufacturing cost is about

\$25/kWh, whereas the selling price of Li-ion batteries is about \$200/kWh. The cycle life of the advanced Fe-air battery using ionic liquid as an electrolyte additive stands at more than 1000 cycles without showing any deterioration in capacity.

Iron-air batteries likely have a theoretical energy density of about 1200Wh/kg, whereas Li-ion batteries come in about 460Wh/kg. When expressed in volumetric energy density, the figure is quite impressive at 9700Wh/l, which is almost five times as high as that of Li-ion batteries (2000Wh/l). The theoretical and practical energy densities of various types of batteries are shown in Fig. 5.

Cell reaction

In Fe-air batteries, most of the products formed during cell reactions are solid, rather than being dissolved in the electrolyte, and there is no dendrite deposition. This makes the fabrication of the battery much simpler, as no separator is required to prevent dissolved metal species from reaching the air electrode.

In many cases, the circulation of electrolyte is not needed inside the cell to ensure the uniform electro-deposition of iron during recharging or to prevent dendrite deposition, except to remove excess heat and replenish the electrolyte. Fe-air cells usually deploy an aqueous KOH solution at 3-6 mol dm⁻³ to ensure good ionic conductivity and enable the reduction and evolution of oxygen reactions easily on the bifunctional catalyst of the oxygen electrode. The solubility of the iron oxide products is very low in an alkaline solution, making the solid-state iron electrode very stable. It is necessary that the catalytic surface of the air electrode is wetted with the electrolyte, but at the same time, the gas diffusion layer remains dry to allow facile oxygen diffusion towards the catalyst layer without any hindrance.

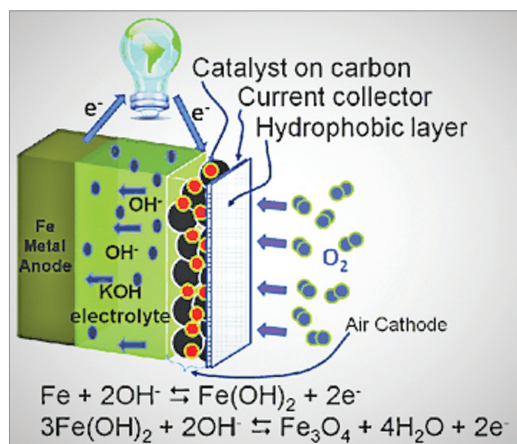


Fig. 4: Schematic arrangement of iron-air battery

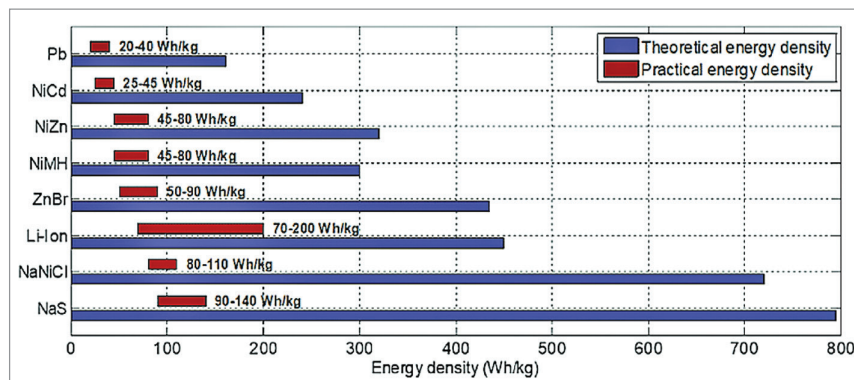


Fig. 5: Theoretical and practical energy density of different batteries